
Beyond Uniform Computation: Spatially-Adaptive Sparse Diffusion Transformers for Efficient Text-to-Image Synthesis

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Abstract

Diffusion Transformers (DiTs) have emerged as the dominant architecture for high-fidelity text-to-image synthesis, yet they suffer from prohibitive computational costs due to their uniform token processing paradigm. This survey examines recent advances in efficient DiT architectures, categorizing approaches into architectural optimizations, token reduction strategies, and hybrid representations. Through critical analysis of state-of-the-art methods including SANA, HART, and Dynamic Token Merging techniques, we identify a fundamental limitation: current approaches fail to exploit the spatial redundancy inherent in visual data. We propose Spatially-Adaptive Sparse Diffusion Transformers (SA-SparseDiT), a novel framework that dynamically allocates compute based on prompt-driven saliency, enabling $3\text{--}5\times$ speedups while maintaining generation quality.

1 Introduction and Motivation

Text-to-image synthesis has undergone a revolutionary transformation with the advent of Diffusion Transformers (DiTs). Models such as Stable Diffusion 3 [1] and FLUX [2] demonstrate unprecedented visual quality by replacing traditional U-Net backbones with scalable transformer architectures. However, this quality comes at a steep computational cost: DiTs process *every* latent token with equal intensity, regardless of semantic importance.

The quadratic complexity $O(N^2)$ of self-attention in DiTs becomes the primary bottleneck when scaling to high resolutions. For a 1024×1024 image with $16\times$ VAE compression, the sequence length $N = 4096$, resulting in $\sim 16\text{M}$ attention operations per layer per timestep. With typical models requiring 50 denoising steps and 20+ transformer layers, total operations exceed 16×10^9 per image generation—prohibitive for real-time applications.

This survey focuses on **Spatially-Adaptive Computation** in DiTs: methods that reduce this burden by selectively processing tokens based on their importance. We examine 8 high-impact papers from CVPR, NeurIPS, and ICLR (2024–2025) that attack this problem from three distinct angles: (1) architectural efficiency, (2) token reduction strategies, and (3) hybrid representations. Our analysis reveals that while recent works achieve impressive speedups, they fail to dynamically align computational allocation with the prompt’s semantic requirements—a critical gap we address in our proposed research direction.

2 Categorization and Taxonomy

We organize the surveyed literature into three complementary paradigms:

2.1 Category A: Architectural Efficiency Improvements

These methods modify the DiT backbone to reduce computational complexity *without* reducing token count, through attention approximations or structural changes.

Papers:

- **SANA 1.5** [3]: Linear Diffusion Transformers with depth-wise convolution attention, reducing complexity from $O(N^2)$ to $O(N)$ while maintaining global receptive fields.
- **EDiT** [4]: Efficient Diffusion Transformers with Linear Compressed Attention using low-rank factorization of attention matrices.

2.2 Category B: Token Reduction and Merging

These approaches explicitly reduce the sequence length N by merging or pruning redundant tokens during inference.

Papers:

- **Token Merging (ToMe)** [5]: Bipartite matching-based merging for Vision Transformers, adapted for diffusion models.
- **ToMA** [6]: Token Merge with Attention, introducing saliency-aware merging strategies specifically for DiTs.
- **Dynamic Chunking DiT** [7]: Recent work introducing patch-level scheduling based on local information density.

2.3 Category C: Hybrid Representations and Alternative Paradigms

These works challenge the pure-diffusion paradigm by combining discrete autoregressive modeling with continuous refinement.

Papers:

- **HART** [8]: Hybrid Autoregressive Transformer combining discrete tokens for structure with continuous residuals for detail, achieving single-pass generation.
- **DDiT** [9]: Dynamic Patch Scheduling using learnable region importance scores.
- **Shiva-DiT** [10]: Residual-Based Differentiable Top- k Selection for token pruning.

Table 1: Taxonomy of Efficient DiT Methods

Method	Category	Speedup	Granularity
SANA 1.5	A: Architectural	2–3×	Global
EDiT	A: Architectural	1.5–2×	Global
ToMe	B: Token Reduction	2×	Spatial (static)
ToMA	B: Token Reduction	2.5×	Spatial (attention-guided)
Dynamic Chunking	B: Token Reduction	3×	Patch-level
Shiva-DiT	B: Token Reduction	2–4×	Token-level (residual)
HART	C: Hybrid	10×	Representation-level

3 Deep Dive and Methodological Comparison

3.1 Architectural Efficiency: The SANA Approach

SANA 1.5 [3] replaces standard self-attention with **Linear Diffusion Transformer** blocks using depth-wise convolutions. The attention mechanism becomes:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \approx \text{DWConv}(Q) \cdot V$$

While this achieves linear complexity $O(N)$, it *uniformly* reduces computation across all tokens. Regions requiring fine detail (faces, text) receive the same “lightweight” attention as simple backgrounds, potentially compromising high-frequency generation.

Trade-off: SANA trades model expressiveness for speed. The linear approximation cannot capture long-range dependencies as effectively as full attention, particularly for complex compositional scenes.

3.2 Token Reduction Strategies: From Static to Dynamic

ToMe (Static Merging). ToMe [5] defines “similarity” via cosine distance in feature space, merging the r most similar token pairs at each layer:

$$\mathcal{M} = \{(i, j) \mid \text{argsort-top-}r(\cos(x_i, x_j))\}$$

Limitation: Feature similarity does not correlate with semantic importance. Two distinct texture patches may have high feature similarity but require independent processing for photorealism.

ToMA (Attention-Guided Merging). ToMA [6] improves upon ToMe by using cross-attention maps $A \in \mathbb{R}^{N \times M}$ (where M is text sequence length):

$$s_i = \sum_{j=1}^M A_{i,j}$$

Tokens with low s_i (poor alignment with text) are prioritized for merging.

Advancement: Text-to-vision alignment improves merging decisions. However, ToMA applies merging *statically* at fixed layers, failing to adapt to the evolving importance of tokens across denoising timesteps.

DDiT and Dynamic Chunking. Recent work [9, 7] introduces temporal adaptivity: token importance is recomputed at each denoising step using learned “Decider” modules or patch-level statistics.

Gap: These methods schedule compute at the *patch* or *chunk* level, not individual tokens. Coarse-grained decisions miss opportunities for fine-grained efficiency.

Shiva-DiT. Shiva-DiT [10] implements differentiable Top- k selection using the Gumbel-Softmax trick:

$$P(\text{keep}_i) = \text{Softmax}\left(\frac{\log \pi_i + g_i}{\tau}\right)$$

where π_i is a learned importance score and $g_i \sim \text{Gumbel}(0, 1)$.

Innovation: End-to-end trainable sparsity. However, the importance score π_i is learned from reconstruction loss alone, lacking explicit *text-prompt guidance*.

3.3 Hybrid Representations: HART

HART [8] abandons the diffusion paradigm entirely for a hybrid autoregressive approach:

1. Discrete VQ-VAE tokens capture global structure in one AR pass.
2. Continuous residuals add detail via small diffusion heads.

Strengths: Achieves single-pass generation (massive speedup), leveraging the efficiency of AR models (GPT-style next-token prediction).

Fundamental Limitation: HART (and all AR models) require *discrete tokenization*, which introduces an information bottleneck at the VQ-VAE stage. Complex textures and continuous gradients

suffer from quantization artifacts. Furthermore, AR models autoregressively decode, making them inherently sequential—batch generation of multiple images cannot be parallelized as effectively as diffusion.

4 Limitations and Open Challenges

Our survey reveals three systematic failures across the state-of-the-art:

4.1 Challenge 1: The Uniformity Assumption

Current approaches assume uniform importance across the spatial domain. Whether applying full attention (standard DiT), linear attention (SANA), or token merging (ToMe), existing methods process the “sky” region with the same intensity as a “human face.”

Why it fails: Visual scenes exhibit extreme spatial redundancy. In most images, 70–80% of pixels are low-entropy (smooth regions, repetitive textures). Current architectures waste compute on these “uninformative” regions while under-investing in high-frequency details critical for perceived quality.

4.2 Challenge 2: Static vs. Dynamic Importance

Text-prompt alignment is underutilized in compute allocation. While ToMA uses cross-attention for merging decisions, it treats this saliency as *constant* across timesteps.

Why it fails: The importance of a token evolves during denoising. At $t = T$ (pure noise), all tokens are equally uncertain. At $t = T/2$, some regions (e.g., backgrounds) may stabilize while others (e.g., foreground objects) remain noisy. Current methods lack mechanisms to dynamically re-prioritize tokens as the diffusion trajectory unfolds.

4.3 Challenge 3: The Differentiable Trilemma

End-to-end trainable sparsity requires balancing three conflicting goals: (1) differentiability (for gradient-based training), (2) efficiency (for wall-clock speedup), and (3) quality preservation (no generation artifacts).

Shiva-DiT achieves (1) and (2) but compromises on (3)—learned importance scores drift toward conservative selection, reducing effective sparsity. Feature caching methods achieve (2) and (3) but are *not* differentiable, requiring hand-tuned heuristics.

Why it matters: Without a unified solution to this trilemma, spatial sparsity remains a “post-hoc” optimization rather than a first-class architectural primitive.

4.4 Challenge 4: Seamless Integration

No existing method handles the boundary between sparse and dense regions. When merging tokens or dropping computation, current approaches risk visible artifacts at region boundaries—the “blocky” effect where high-detail subjects meet low-detail backgrounds.

Real-world impact: These limitations prevent deployment of high-quality DiTs on consumer hardware. Current models require 24GB+ VRAM and 10–30 seconds per image, limiting applications in real-time creative tools, mobile AR, and autonomous systems requiring rapid visual imagination.

5 Proposed Research Direction (Towards a NeurIPS Submission)

5.1 The Specific Limitation to Solve

We target **Challenge 2 (Dynamic Importance)** and **Challenge 4 (Seamless Integration)**. Our hypothesis is that *compute allocation should be a function of both the text prompt (semantic saliency) and the current timestep (temporal stability)*.

5.2 Proposed Solution: SA-SparseDiT

We propose **Spatially-Adaptive Sparse Diffusion Transformers (SA-SparseDiT)**, a DiT variant with three novel components:

5.2.1 1. Tri-Signal Saliency Router

A differentiable gating module that computes per-token priority P_i using three signals:

$$P_i = \underbrace{\alpha \cdot A_{\text{text},i}}_{\text{Text Alignment}} + \underbrace{\beta \cdot \|\nabla_{\theta} \epsilon_i\|_2}_{\text{Noise Magnitude}} + \underbrace{\gamma \cdot \mathcal{V}(x_i)}_{\text{Spatial Variance}}$$

where:

- $A_{\text{text},i}$: Cross-attention weight to text embeddings (semantic importance).
- $\|\nabla_{\epsilon}\|_2$: Magnitude of predicted noise (temporal uncertainty).
- $\mathcal{V}(x_i)$: Local spatial variance (structural complexity).

Why this is novel: Unlike ToMA (static attention) or Shiva-DiT (residual only), our router dynamically recomputes priorities at each denoising step, allowing tokens to “graduate” from sparse to dense processing as they become semantically relevant.

5.2.2 2. Differentiable Top- k Selection via Gumbel-Softmax

Using the concrete distribution [11], we implement sparse selection:

$$z_i = \text{sigmoid}\left(\frac{\log P_i + g_i}{\tau}\right), \quad g_i \sim \text{Gumbel}(0, 1)$$

Top- k tokens ($z_i > \tau_{\text{sparsity}}$) proceed through full self-attention. **Bottom- $(N-k)$ tokens** are merged into $m \ll k$ anchor tokens via a learned clustering operation.

5.2.3 3. Seamless Re-Integration via Residual Flows

To address Challenge 4, we introduce **seamless un-pooling** using normalizing flows [12]. When expanding merged tokens back to full resolution, we apply invertible transformations that preserve local smoothness while allowing high-frequency detail injection from dense regions.

5.3 Expected Contributions

1. **First prompt-driven dynamic sparsity** for DiTs with per-token, per-timestep re-prioritization.
2. **Differentiable sparse attention kernels** enabling end-to-end training without quality degradation.
3. **3–5× speedup** on 1024p generation with $< 5\%$ FID increase, enabling 8K synthesis on consumer GPUs.

5.4 Risk Mitigation

- **Rebuttal: “HART already solves efficiency.”** → HART requires discrete tokenization. SA-SparseDiT preserves the continuous flow-matching paradigm, maintaining superior texture quality.
- **Rebuttal: “Dynamic Chunking DDiT already token-prunes.”** → DDiT operates at patch-level (16×16 blocks). SA-SparseDiT is token-level, offering finer granularity.

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